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RESEARCH ARTICLE

Feasibility of PV-Based Islanded Microgrids for Affordable Electricity in Sub-Saharan Africa

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ABSTRACT

Many countries in Sub-Saharan Africa (SSA) suffer from the lack of access to electricity due to poverty and a dispersed population. Due to the solar power potential in this region, solar-powered microgrids were examined in this study as a way to supply affordable electricity to the rural population in SSA. Using HOMER Pro as the microgrid optimization tool, a microgrid designed to supply electricity in Burkina Faso was simulated. With one of the cheapest electricity tariffs in SSA ($0.240/kWh), Burkina Faso was selected to ensure that the simulation results can serve as a proof-of-concept for all of the Sub-Saharan countries. Factors like solar panel output degradation and dusting, unexpected component failure, and linearity of the load profile were assessed to determine the lowest-cost and highest-cost results for the simulated microgrid. The performed simulations showed that even without a connection to the main grid, a solar-powered microgrid can be easily profitable while supplying electricity at below the average national tariff, with almost no expected power outages. This paper proves that microgrids with only solar panels as a power source remain a viable option to provide cheap electricity to the impoverished rural regions in SSA, even with almost no component maintenance.

Index Terms—Cost of electricity, electricity access, HOMER pro, microgrid, solar power, Sub-Saharan Africa.

I. INTRODUCTION

A microgrid is an autonomous energy system that supplies electricity to a region by using several power generation methods. Microgrids can function independently; if they are connected to the main grid, they can isolate themselves in case of an emergency. A microgrid that maintains no connection to the main grid is called an islanded microgrid. Islanded microgrids are principally used for providing electricity to isolated villages or outlying regions. Generally, connecting faraway places to the main grid is extremely costly; therefore, a microgrid remains a preferable option. In an islanded configuration, the power generation must be adequate at all times to constantly supply the microgrid's sensitive loads, in accordance with IEEE Std 1547.4-2011 [1]. For this reason, batteries are used to continuously provide electricity when power generation becomes insufficient [2].

For African countries that have large rural regions suffering from poverty, and have no working national grids to supply them with electricity, islanded microgrids become the sole option. The World Bank SE4ALL database's rural population density data are critically valuable in this regard. The data suggest that there are more isolated rural populations in Africa than in the rest of the world [3]. Furthermore, most of the countries in Africa are poverty-stricken. According to the World Bank's World Development Indicator report in 2017, the share of the population that lives in extreme poverty (income of <$1.9 per day) is slightly below 50% in Sub-Saharan Africa (SSA) [3]. The statistics show that SSA, in particular, has exhibited signs of extreme poverty, which explains the region's lack of electricity access. Electricity access in SSA is only 30.5% overall. Rural access is worse: 14.2% [4]. The result is higher electricity tariffs in SSA, which ranges between $0.200/kWh and $0.500/kWh, depending on the country [5]. The worldwide electricity tariff average is $0.134/kWh in comparison [6].

This paper aims to prove that providing cheaper electricity in SSA is possible with photovoltaic (PV)-based islanded microgrids. For this reason, a PV-based microgrid was planned in Burkina Faso, owing to the cheaper overall electricity tariff than other Sub-Saharan countries [7]. Being able to provide cheaper electricity in Burkina Faso ensures that this study can be applicable to most, if not all, of the Sub-Saharan countries in terms of providing cheaper electricity to

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the population. The simulation was performed using HOMER Pro, a microgrid design and optimization tool.

The rest of the paper includes the following:

- Location selection for the microgrid project with detailed justification for the selection.
- Explanation of the statistical analysis used by HOMER Pro to simulate the microgrid.
- Comprehensive analysis for component selection.
- Presentation of the results.
- Information about the limitations that affected the results.
- Comments about the findings in this paper and their implications.

II. METHODS

A. Location Selection

Burkina Faso's low electrification rates demonstrate the need for a microgrid project. With a combined population of 19.2 million [3], Burkina Faso's urban population makes up 29% of the total population, and only 58% of that urban population has electricity access. Moreover, only 3% of the rural population has electricity access. Note that 70% of Burkina Faso's population lives in rural areas [8]. In addition, Burkina Faso's inability to seamlessly supply electricity to regions under the national grid reinforces the need for the project [9].

Province selection was influenced by several factors. The solar irradiation map of Burkina Faso in [7] was used to determine candidate regions where a solar-powered microgrid could provide maximum power. The referenced map revealed that the country's Northeast region is a suitable region for the microgrid. The next step was to take the population distribution for the suitable regions into account, to identify the most suitable location for the microgrid. The grid also needs to be relatively close to the main grid to enable a possible future main grid connection. After comparing the population centers and grid distribution with the solar irradiation map, the Namentenga province was selected [7]. With a population of 476 262, this province is relatively populous. Notably, 95.3% of the Namentenga province's population lives in rural areas, where only 1% of the population on average has access to electricity [10]. The exact location of the microgrid project is 13°35.9'N, 0°16.5'W, which lies inside the Department of Yalgo.

Main Points

- HOMER Pro is used as the microgrid optimization tool to determine the most affordable solutions for the simulated microgrid in Burkina Faso. The pros and cons of HOMER Pro are discussed.
- The effects of maintenance and external factors such as solar panel dusting are evaluated by simulating different lifetime expectations for each component in the microgrid and different derating factor possibilities for the solar panels.
- The importance of load profile linearization is discussed by proving that a more linearized load profile results in a lower cost of electricity (COE).
- It is shown that even with almost no component maintenance, the PV-based microgrid solution offers a COE that can be profitable while setting an electricity tariff that is lower than Burkina Faso's average tariff.
- A future connection to the main grid can lower the microgrid's COE further by selling the excess energy that is wasted while the microgrid is in islanded configuration.

B. Statistical Analysis Using HOMER Pro

For the designed microgrid, multiple simulations for multiple scenarios are performed to establish the most economical solution. The project location is used to get the monthly average solar irradiance and clearness index data from HOMER Pro. Clearness index (K_T) is defined as the ratio of the solar radiation that reaches the Earth's surface. This value is between 0 and 1. The clearness index is closer to 1 on a sunny and clear day, whereas it gets closer to 0 in cloudy conditions. The calculation of the clearness index is performed with (1), where H_{ave} is the monthly average radiation on the horizontal surface of the Earth and $H_{o,ave}$ is the extraterrestrial horizontal radiation. In other words, $H_{o,ave}$ is the radiation on a horizontal surface at the top of the Earth's atmosphere [11];

$$K_T = \frac{H_{ave}}{H_{o,ave}} \tag{1}$$

With a given H_{ave}, HOMER Pro can find K_T after calculating $H_{o,ave}$, which involves several steps. The initial step is to calculate the intensity of solar radiation at the top of the Earth's atmosphere (G_{on}) (2);

$$G_{on} = G_{sc} \cdot \left(1 + 0.033 \cdot \cos\left(\frac{360 \cdot n}{365}\right)\right) \tag{2}$$

G_{sc} is the solar constant (1.367 kW/m^2), whereas n is the day of the year. The next step is to obtain the extraterrestrial radiation on the horizontal surface (G_o) which changes with the zenith angle (θ_z). The zenith angle represents the angle between the sun's rays and the vertical. G_o can be calculated using (3) and (4):

$$G_o = G_{on} \cdot \cos\theta_z \tag{3}$$

$$\cos\theta_z = \cos\phi \cdot \cos\delta \cdot \cos\omega + \sin\phi \cdot \sin\delta \tag{4}$$

For (4), ϕ is the latitude, δ is the solar declination, and ω is the hour angle. The solar declination itself is calculated using (5):

$$\delta = 23.45° \cdot \sin\left(360° \cdot \left(\frac{284 + n}{365}\right)\right) \tag{5}$$

A further step involves finding H_o, the average daily extraterrestrial radiation per square meter. Integrating (3) from sunrise to sunset will give H_o. The result of the integration is given in (6):

$$H_o = \frac{24}{\pi} \cdot G_{on} \cdot \left(\cos\phi \cdot \cos\delta \cdot \sin\omega_s + \frac{\pi \cdot \omega_s}{180°} \cdot \sin\phi \cdot \sin\delta\right) \tag{6}$$

ω_s is the sunset hour angle, which can be calculated using (7):

$$\cos\omega_s = -\tan\phi \cdot \tan\delta \tag{7}$$

Finally, $H_{o,ave}$ can be calculated (8). N is the number of days in the month:

$$H_{o,ave} = \frac{\sum_{n=1}^{N} H_o}{N} \tag{8}$$

Using the provided equations, the monthly average solar global horizontal irradiance (GHI) data for Namentenga are presented in Table I.

One of HOMER Pro's most important features is its ability to conduct accurate economic analysis. Attaining accurate results requires data about several economic properties of the region. These properties include the country's inflation rate and nominal discount rate. The nominal discount rate is the interest rate without the effects of inflation. Both inflation and nominal discount rates are used to calculate the cost increase, caused by inflation, of any materials that need to be replaced. Maintenance costs are equally affected by inflation: calculating the maintenance costs of a project with a lifetime of multiple years cannot be done correctly without considering the effects of inflation in the region. Over the last 18 years, Burkina Faso's average inflation rate (*f*) has been 2.3%, whereas the nominal discount rate (*i'*) is 5.14% [12, 13]. These rates are used to determine the real discount rate (real interest rate). The real discount rate (*i*) is the interest rate used to remove the effect of inflation, which is:

$$i = \frac{i' - f}{1 + f} \tag{9}$$

Using (9), the real discount rate is calculated to be 2.78%.

HOMER Pro can also calculate the COE and total net present cost (NPC). The COE is the average cost of producing 1 kWh of electrical energy by the system. NPC represents the entire project's cost for the full length of its determined lifetime. For COE and NPC, the goal is to keep those costs as low as possible. Using (10), COE can be calculated:

$$COE = \frac{Cyear}{E(AC) + E(DC)} \tag{10}$$

C_{year} is the total annualized cost of the microgrid, whereas *E(AC)* and *E(DC)* are the yearly spent AC and DC loads, respectively.

Determining the NPC requires calculating the capital recovery factor (CRF), which is a ratio used to calculate the present value of the cash flow. The present value of a payment changes in time. This change is dependent on the real discount rate and the amount of time, as is apparent with the CRF's following formula (11):

$$CRF(i,N) = i \cdot \frac{(1+i)^N}{(1+i)^N - 1} \tag{11}$$

Note that *N* is the project lifetime, which is 25 years for the project in this paper. Using CRF, NPC can be calculated as (12):

$$NPC = \frac{Cyear}{CRF} \tag{12}$$

After the project lifetime is reached, every component is sold as salvage to recuperate some of the installation costs. The salvage value is calculated using (13):

$$S = C_{rep} \cdot \frac{R_{rem}}{R_{comp}} \tag{13}$$

C_{rep} is the replacement cost, R_{rem} is the remaining lifetime at the end of the project lifetime, and R_{comp} is the component lifetime, which is:

$$R_{rem} = R_{comp} - (N - R_{rep}) \tag{14}$$

R_{rep} is the replacement cost duration, calculated by (15):

$$R_{rep} = R_{comp} \cdot INT\left(\frac{R_{proj}}{R_{comp}}\right) \tag{15}$$

TABLE I
SOLAR GLOBAL HORIZONTAL IRRADIANCE (GHI) DATA FOR NAMENTENGA REGION

Month	**K_T**	**H_{ave} (kWh/m²/day)**	**Month**	**K_T**	**H_{ave} (kWh/m²/day)**
January	0.638	5.360	July	0.573	6.050
February	0.680	6.260	August	0.539	5.670
March	0.666	6.690	September	0.573	5.830
April	0.650	6.860	October	0.625	5.880
May	0.638	6.780	November	0.648	5.530
June	0.613	6.470	December	0.632	5.130

K_T, clearness index; H_{ave}, monthly average radiation.

Note that *INT ()* gives the integer value of the equation inside the parentheses.

One can also limit the annual capacity shortage to make sure that the project sufficiently provides consistent electricity to its users. HOMER Pro assumes an annual capacity shortage of 0% by default. However, this can result in situations where a specific load may require excessive energy production to ensure no capacity shortage during peak loads. The result is wasted energy and an increase in the COE and NPC if the microgrid is islanded. After several preliminary simulations, an annual capacity shortage of 2% was allowed in the system.

Furthermore, the project's load profile needs to be determined. The software has certain artificial load profiles and multiple load profiles for several locations within the United States. These load profiles can aid a project even if it is outside the United States by only listing load profiles that match the climate of the project's location. Because electrification of a rural region is desired for the microgrid design, a residential load profile can be assumed. However, the load profile cannot be expected to remain the same throughout the project's 25-year lifespan. Once the electrification is achieved and businesses start to establish in the area, the load profile will turn into a more distributed one. The reason is that the electricity usage will be more varied instead of electricity used primarily for illumination at night and refrigeration. Investigating this factor requires one simulation each for two different load profiles (residential and community, Fig. 1 and Fig. 2). From the results, a two-step microgrid installation was planned. The first step was assumed for a complete residential load profile, serving a small fraction of the planned amount, owing to the less linear nature of the load profile increasing the COE. The second step was assumed for a community load profile. To find the best annual average load for each step, the scale of the project for each load profile needs to be examined. For this reason, a load that satisfies the needs of between 500 and 5000 people was considered, with an increment of 500 people. Yearly average energy consumption per capita in Burkina Faso is 76 kWh [14], which is equal to 208.22 Wh/day. Thus, the average annual load in the simulation is in the range of 104.11–1041.1 kWh/day. Note that with a community load profile, average energy consumption per capita in the region might increase owing to increased economic activity.

C. Component Selection

The required components for the microgrid are solar panels for power generation, maximum power-point (MPP) tracking (MPPT) charge controllers to always output maximum power at a stable DC bus voltage [15], batteries for energy storage, and inverters for DC–AC conversion. There are several considerations to be made for component selection:

- DC bus voltage needs to be at the DC voltage range of the inverter. In addition, most MPPT charge controllers depend upon solar panel output voltage to be above the battery voltage to function. Thus, DC bus voltage needs to be selected in accordance with this fact.
- Battery nominal voltage needs to be at the required DC bus voltage. The string size can be altered to accommodate this specification.
- The selected MPPT charge controller needs to be able to charge the batteries at the specified DC bus voltage. It also needs to

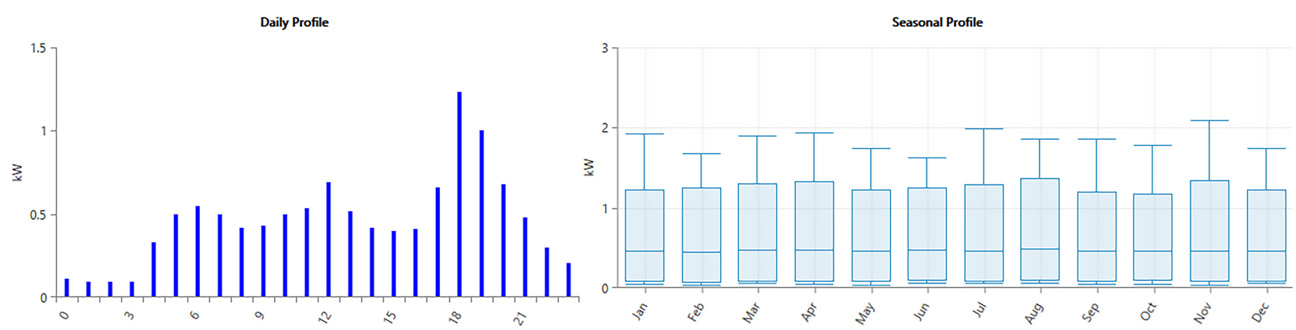

Fig. 1. Residential daily and seasonal load profile.

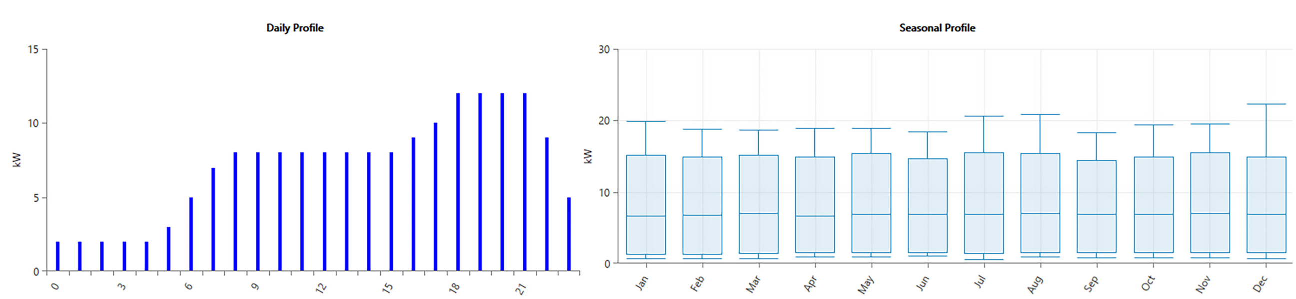

Fig. 2. Community daily and seasonal load profile.

be rated at appropriate maximum PV short circuit current (I_{sc}) and open circuit voltage (V_{oc}) values. A solar panel with a higher short circuit current can damage the controller. The datasheet for the MPPT controller needs to be examined for more potential restrictions.

For solar panels, LG Electronics 365Q1C-A5 was selected for its high efficiency (21%). According to the panel's datasheet, its maximum power (P_{max}) is 365 W, V_{oc} is 42.8 V, and I_{sc} is 10.8 A. Moreover, the output of the solar panel is expected to decline by 2% during the first year. Afterward, an annual output decline of 0.4% is expected [16]. Based on this information, the average solar panel output after 25 years is 93.46% of its initial performance. This factor is included in the simulation as the derating factor (f_{PV}) of the solar panels.

The MPP voltage of the solar panels is 36.7 V [16]. This necessitates a DC bus voltage lower than this value to make sure that the MPPT charge controller is functional. Because most MPPT charge controllers support battery voltages of 12–24–36–48 V, a 12 V DC bus voltage is selected for this project to ensure the continuous operation of MPPT charge controllers.

For energy storage, EnerSys PowerSafe SBS 900 batteries were used. They have a nominal voltage of 12 V and a nominal capacity of 12.1 kWh. The string size does not need to be altered because both the DC bus voltage and the battery nominal voltage are the same. These batteries also need to undergo maintenance every six months. This requirement was factored into the simulation.

The selected MPPT charge controller is Victron BlueSolar MPPT 100/50. Its rated charge current is 50 A; therefore, it is capable of providing 700 W of nominal power at 12 V battery voltage. Its specified maximum V_{oc} and I_{sc} values are 100 V and 60 A, respectively, which satisfy the solar panels' specifications. For this system, each MPPT charge controller can handle two solar panels connected in series. Even though the combined P_{max} is 730 W, the simulation assumes 93.46% of maximum solar panel output, because of the output power degradation mentioned earlier. For this reason, the actual P_{max} is slightly above 682 W, which the MPPT charge controller can provide [17].

Studer Xtender XTH 3000-12 was selected for DC–AC conversion. Its nominal battery voltage is 12 V and it has an input voltage range between 9.5 and 17 V. It can supply continuous power of 2500 VA [18].

Each selected component needs to be replaced after several years, owing to aging. Each component has a certain lifetime specified both in the respective datasheets and in the HOMER Pro software. However, the need for replacement can occur earlier than expected because of suboptimal storage conditions and lack of maintenance. For this reason, several lifetime possibilities were specified for each component. A lack of maintenance can also lower solar panel power output owing to dusting, especially in Africa [19]. This reduction varies between 4% and 32%, according to results in [20–29]. To study the most undesirable possible outcome, the power output reduction of 32% was examined with another derating factor: 63.55% (32% less than the average output of the solar panels throughout their 25-year lifespan). The considered lifetimes, alongside the capital, and operation-and-maintenance (O&M) costs are presented in Table II. Multiple technical reports and papers were used to decide the capital and O&M costs [30–34].

III. RESULTS

Owing to the many sensitivity issues due to the components' lifetime considerations and a change in the f_{PV} of the solar panels, results are provided in steps. These steps are as follows:

- The best case scenario for each average annual load is presented. The load with best results is selected.
- For the selected load, the effect of components' lifetime is provided. f_{PV} is 93.46% for the given results.
- The effect of dusting on solar panels owing to lack of maintenance is then examined using f_{PV} as another sensitivity issue.

A. Simulation Results for the Residential Load Profile

The results for the residential load profile with the best replacement requirements for each component are presented in Table III. Load #3 has the lowest COE of $0.146/kWh. Considering the fact that the average COE in Burkina Faso is $0.194/kWh, this result is definitely acceptable as it allows setting a tariff much lower than Burkina Faso's average ($0.240/kWh) [7]. Lowering electricity tariffs in Burkina Faso is essential, considering the prevalent poverty [3]. In addition, load #3 also provides the best autonomy among all other options. Note that autonomy refers to the amount of time that the microgrid can be sustained by the power accumulated in the batteries, in case of a blackout or during maintenance operations.

TABLE II
COMPONENTS SELECTED FOR THE MICROGRID AND THEIR ASSOCIATED COSTS

Component Type	Component Name	Cost ($/unit)	Simulated Lifetime (years)	O&M Costs ($/unit-year)
Solar Panel	LG 365Q1C-A5	380.00	25, 20	10.95
MPPT Charge Controller	Victron BlueSolar MPPT 100/50	343.00	20, 15	Included in solar panel O&M costs
Battery	EnerSys PowerSafe SBS 900	1440.00	15, 10	25.00
Inverter	Studer Xtender XTH 3000-12	901.00	10, 5	100.00

TABLE III
BEST RESULTS FOR EACH ANNUAL AVERAGE LOAD (RESIDENTIAL LOAD PROFILE)

#	Annual Average Load (kWh/day)	NPC ($)	COE ($/kWh)	Operating Cost ($/year)	Initial Capital ($)	Autonomy (hours)	Excess Electricity (%)
1	104.11	99 240	0.148	2 497	54 650	43.0	10.40
2	208.22	196 958	0.147	4 889	109 667	44.0	10.80
3	312.33	295 067	0.146	7 040	169 363	49.5	11.80
4	416.44	396 499	0.148	9 863	220 390	42.5	12.50
5	520.55	494 683	0.147	12 158	277 597	45.4	11.70
6	624.66	595 689	0.148	15 021	327 479	39.4	13.40
7	728.77	701 454	0.150	17 921	381 473	39.9	11.90
8	832.88	793 141	0.148	19 624	442 739	41.8	14.00
9	936.99	891 060	0.147	21 977	498 642	41.1	14.90
10	1041.1	995 415	0.148	25 197	545 503	39.5	13.10

NPC, net present cost; COE, cost of electricity.

TABLE IV
EFFECT OF COMPONENTS' LIFETIME FOR LOAD #3 (F_{PV} = 93.46%)

#	Component Lifetime (years)				COE ($/kWh)	NPC ($)	Autonomy (hours)	Excess Electricity (%)
	N_{Inv}	N_{Bat}	N_{PV}	N_{MPPT}				
3.1	10	10	20	10	0.176	354 746	40.4	12.10
3.2	10	15	20	10	0.169	341 742	48.9	11.80
3.3	10	10	25	10	0.169	339 684	40.4	12.10
3.4	10	15	25	10	0.162	325 889	45.0	11.70
3.5	5	10	20	10	0.190	383 676	41.7	12.40
3.6	5	15	20	10	0.183	368 215	44.3	12.90
3.7	5	10	25	10	0.184	370 756	43.0	12.00
3.8	5	15	25	10	0.175	352 123	50.2	11.90
3.9	10	10	20	20	0.160	322 700	37.8	14.10
3.10	10	15	20	20	0.154	310 102	49.5	11.80
3.11	10	10	25	20	0.153	307 133	37.2	15.00
3.12	10	15	25	20	0.146	295 067	49.5	11.80
3.13	5	10	20	20	0.176	354 418	43.0	12.00
3.14	5	15	20	20	0.167	336 082	47.6	12.30
3.15	5	10	25	20	0.167	336 463	37.8	15.20
3.16	5	15	25	20	0.159	320 511	46.9	11.70

NPC, net present cost; COE, cost of electricity.

TABLE V
BEST RESULTS FOR EACH ANNUAL AVERAGE LOAD (COMMUNITY LOAD PROFILE)

#	Annual Average Load (kWh/day)	NPC ($)	COE ($/kWh)	Operating Cost ($/year)	Initial Capital ($)	Autonomy (hours)	Excess Electricity (%)
11	104.11	97 975	0.146	2 494	53 445	39.1	10.20
12	208.22	183 292	0.136	4 394	104 828	39.1	11.90
13	312.33	276 256	0.137	6 448	161 129	44.3	11.60
14	416.44	367 460	0.137	8 762	211 003	40.6	11.70
15	520.55	467 017	0.140	11 146	267 990	44.6	9.93
16	624.66	556 652	0.138	13 127	322 253	39.4	15.10
17	728.77	644 890	0.137	15 466	368 738	40.5	11.30
18	832.88	747 539	0.139	18 298	420 818	35.7	14.30
19	936.99	823 816	0.136	19 596	473 910	41.1	11.20
20	1041.1	918 480	0.137	21 679	531 397	41.5	12.20

NPC, net present cost; COE, cost of electricity.

The effect of nonideal replacement requirements for load #3 is examined in Table IV. Even the worst scenario (load #3.5) has a COE ($0.190/kWh) lower than Burkina Faso's average COE. The worst possible dusting further increases the COE to $0.239/kWh, which leaves almost no room for profit without going above the electricity tariff average in Burkina Faso. Note that this is the absolute worst possible result for load #3, owing to the lack of maintenance. Moreover, the solar panels' lifetime typically ranges between 25 and 40 years [35], which makes a COE of $0.239/kWh unlikely for this project.

B. Simulation Results for the Community Load Profile

After an increased economic activity during the day is achieved in the area, a more linearized load profile is expected. Owing to load profile linearization, the relative peak load is reduced, therefore costs are reduced as well, as shown in Table V. The results are achieved with the advertised lifetime of each component. The highest average annual load with the lowest COE of $0.136/kWh is load #19. A comparison between Tables III and V emphasizes the importance of load profile linearization: with a community load profile, a decrease in the COE, NPC, initial capital, and yearly operating costs are achieved in every considered annual average load. In addition, tariffs can be set even lower than what was possible with a residential load profile. Compared with load #3, though, the autonomy decreased from 49.5 hours to 41.1 hours, which can be alleviated by further battery acquisition, but with the caveat of increased COE. As a side note, it must be emphasized that even the COE for load #19 is higher than the worldwide average tariff ($0.134/kWh). This is because of the excess electricity. Thus, the proximity of the main grid to the selected location was crucial owing to this problem: with a future integration to the main grid, excess electricity can be sold to the main grid, thereby reducing the overall COE for the microgrid itself.

Table VI shows the effect of different component lifetimes for load #19. The worst scenario is given in load #19.6. Even though load #19.5 is identical with load #19.6, with the exception of a worse battery lifetime (10 years), it has a lower COE. Owing to the fact that batteries can last longer in load #19.6, it is more sensible to invest more in them, increasing autonomy. However, the subsequent need for more inverters actually increased the COE of load #19.6, which is $0.179/kWh with no PV dusting and $0.221/kWh with PV dusting. The results in Table VI reinforce the fact that a PV-based microgrid project is feasible in Burkina Faso.

IV. DISCUSSION

Several limitations were present owing to the available information about Burkina Faso and the HOMER Pro software. These limitations are as follows:

- No credible information is available about land costs in Burkina Faso. This information is required for more accurate NPC calculation.
- Implementation of the derating factor (f_{PV}) as a way for simulating real-world operating conditions in HOMER Pro is limited because there is no way to input annual power output degradation for solar panels by setting different values for f_{PV} for each year. Entering an f_{PV} for the solar panels' average 25-year output was the best available, albeit still flawed, solution for this limitation. The flaw of this approach is apparent when $N_{PV} = 20$ years, where the solar panels' average output would be higher than 93.46%, especially for the replacements. The simulation of solar panel dusting for this project was also simplistic, owing to the aforementioned limitation of the software.
- No credible information is available about load profiles in the rural regions of Burkina Faso, owing to the lack of electricity access in the country. More accurate results can be achieved

TABLE VI
EFFECT OF COMPONENTS' LIFETIME FOR LOAD #19 (F_{PV}=93.46%)

#	Component Lifetime (years)				COE ($/kWh)	NPC ($)	Autonomy (hours)	Excess Electricity (%)
	N_{Inv}	N_{Bat}	N_{PV}	N_{MPPT}				
19.1	10	10	20	10	0.166	1 002 603	37.2	14.00
19.2	10	15	20	10	0.171	1 031 436	39.1	10.60
19.3	10	10	25	10	0.158	956 404	36.9	14.30
19.4	10	15	25	10	0.158	950 075	32.6	16.40
19.5	5	10	20	10	0.176	1 063 931	36.9	14.30
19.6	5	15	20	10	0.179	1 079 750	39.4	10.40
19.7	5	10	25	10	0.168	1 017 562	36.9	14.30
19.8	5	15	25	10	0.172	1 035 580	39.8	10.40
19.9	10	10	20	20	0.150	903 103	35.6	14.30
19.10	10	15	20	20	0.155	934 489	34.8	13.10
19.11	10	10	25	20	0.142	856 817	35.6	14.30
19.12	10	15	25	20	0.136	823 816	41.1	11.20
19.13	5	10	20	20	0.160	967 227	36.9	14.30
19.14	5	15	20	20	0.155	936 235	38.2	12.60
19.15	5	10	25	20	0.153	922 850	37.4	14.30
19.16	5	15	25	20	0.146	885 176	45.6	11.80

NPC, net present cost; COE, cost of electricity.

with a more location-specific load profile; however, the load profiles used for the project in this paper are accurate enough to base conclusions on.

- Lack of power flow analysis in HOMER Pro makes the optimization fairly simplistic: stability issues that typically arise with intermittent power generation cannot be simulated, although sudden increase in load can be taken into account by setting a day-to-day random variability of the load profile. Furthermore, line losses are also disregarded.

In conclusion, it was found that even without a grid connection, a PV-based microgrid remains a robust option for providing more affordable electricity to Burkina Faso, a country with a lower average electricity tariff than other SSA countries. Component maintenance was proven to be a crucial factor in making sure that the COE remains low throughout the microgrid's operational lifetime. The linearity of the load profile was also shown to affect the results. However, a future main grid connection is required to use the more than 10% excess electricity generated in almost all of the loads in Tables III, IV, V, and VI. This would also decrease the COE, to a level lower than the worldwide average electricity tariff.

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